

Lem transformada fourier gaussiana

Lema 1. Sea g dada por $\forall x \in \mathbb{R} : g(x) = e^{-\pi x^2}$, entonces $g^\wedge = g$.

Demostración: Observamos que $g \in \mathcal{L}^1(\mathbb{R})$ y $\int_{\mathbb{R}} g(x) \, dx = 1$. Calculamos:

$$\begin{aligned} g^\wedge(\xi) &= \int_{\mathbb{R}} e^{-\pi x^2} e^{-2\pi i x \xi} \, dx = \int_{\mathbb{R}} e^{-\pi(x^2 + 2ix\xi)} \, dx = \int_{\mathbb{R}} e^{-\pi(x+i\xi)^2} e^{-\pi\xi^2} \, dx \\ &= e^{-\pi\xi^2} \int_{\mathbb{R}} e^{-\pi(x+i\xi)^2} \, dx = e^{-\pi\xi^2} \int_{\mathbb{R}} e^{-\pi y^2} \, dy = e^{-\pi\xi^2} \cdot 1 = g(\xi). \end{aligned}$$

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